POLITECHNIKA GDAŃSKA

Maszyny cieplne z perspektywy mechaniki kwantowej

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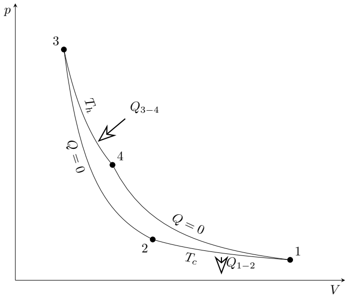
promotor: dr Piotr Weber

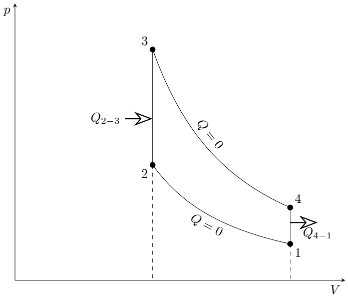
Spis treści

Cel pracy

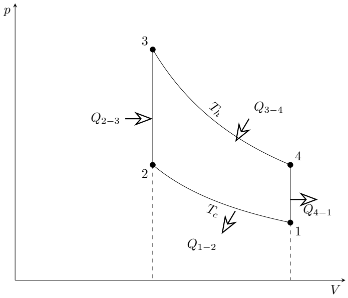
- Przedstawienie teorii termodynamiki kwantowych silników cieplnych
- Wyprowadzenie wielkości termodynamicznych opisujących wybrane silniki kwantowe
- Optymalizacja parametrów silnika



Silnik Carnota



Silnik Otto



Silnik Stirlinga

Silnik Carnota

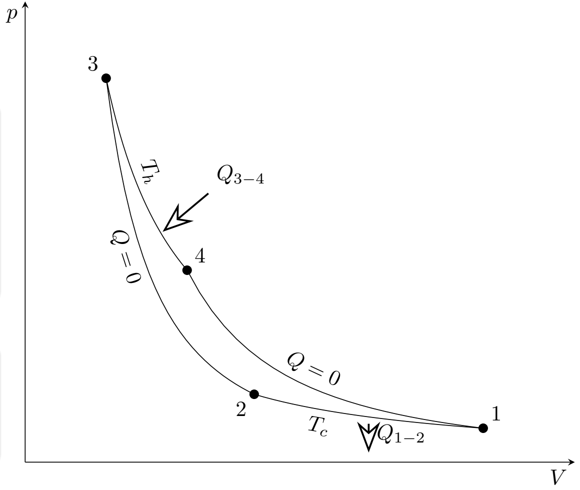
Ciepła

$$Q_{1-2} = nRT_c \ln\left(\frac{V_2}{V_1}\right),$$

$$Q_{3-4} = nRT_h \ln\left(\frac{V_4}{V_3}\right)$$

Sprawność silnika

$$\eta = 1 - \frac{T_c}{T_h}$$



Schemat silnika Carnota

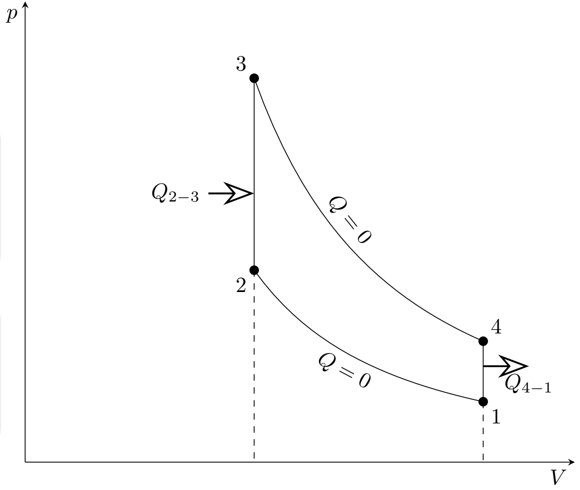
Silnik Otto

Ciepła

$$Q_{2-3} = nc_V(T_3 - T_2),$$
$$Q_{4-1} = nc_V(T_1 - T_4)$$

Sprawność silnika

$$\eta = 1 - \frac{T_4 - T_1}{T_3 - T_2} = 1 - \left(\frac{V_2}{V_1}\right)^{\varkappa+1}$$



Schemat silnika Otto

Silnik Stirlinga

Ciepła

$$Q_{1-2} = nRT_c \ln\left(\frac{V_2}{V_1}\right),$$

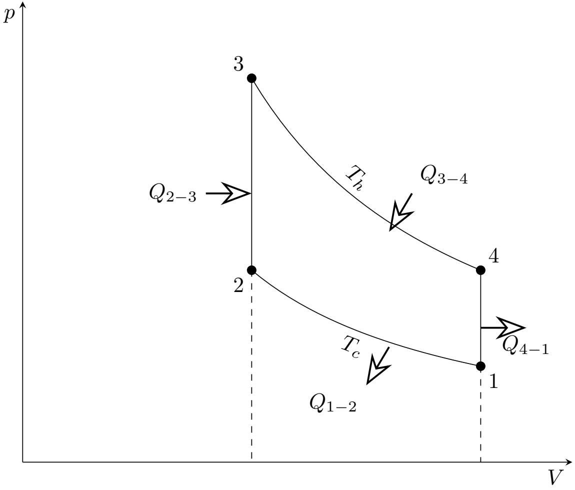
$$Q_{2-3} = nc_V(T_h - T_c),$$

$$Q_{3-4} = nRT_h \ln\left(\frac{V_1}{V_2}\right),$$

$$Q_{4-1} = nc_V(T_c - T_h)$$

Sprawność silnika

$$\eta = 1 - \frac{T_c}{T_h}$$

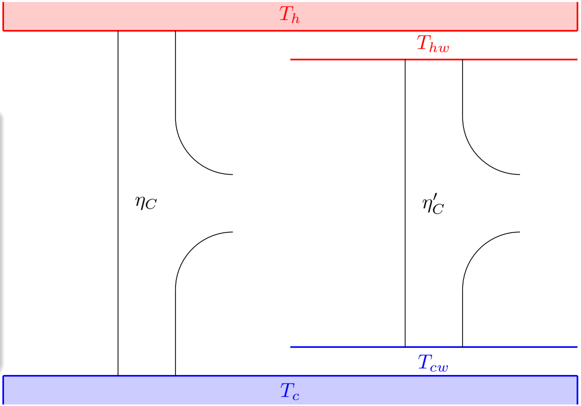


Schemat silnika Stirlinga

Warunki endoodwracalne

- Równowaga wewnętrzna
- Nierównowaga z otoczeniem

$$x = T_h - T_{hw},$$
$$y = T_{cw} - T_c,$$
$$\dot{Q}_h \propto x,$$
$$\dot{Q}_c \propto y$$



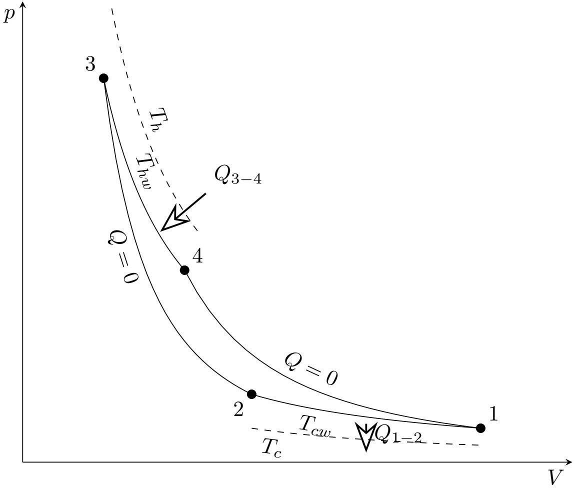
Schemat warunków endoodwracalnych

Endoodwracalny silnik Carnota

Przemiany izotermiczne

$$Q_{1-2} = \alpha_c \tau_c (T_c - T_{cw}) = -\alpha_c \tau_c y,$$
$$Q_{3-4} = \alpha_h \tau_h (T_h - T_{hw}) = \alpha_h \tau_h x.$$

$$\frac{Q_{1-2}}{T_{cw}} = -\frac{Q_{3-4}}{T_{hw}},$$
$$\frac{\tau_h}{\tau_c} = \frac{\alpha_c y (T_h - x)}{\alpha_h x (T_c + y)}$$



Schemat silnika Carnota

Endoodwracalny silnik Carnota

Moc silnika

$$P(x,y) = \frac{\alpha_h \alpha_c x y}{\zeta} \frac{T_h - T_c - x - y}{\alpha_h T_c x + \alpha_c T_h y + (\alpha_h - \alpha_c) x y}$$

Warunki maksimum

$$\frac{\partial P}{\partial x} = 0, \qquad \frac{\partial P}{\partial y} = 0,$$

$$\frac{\partial^2 P}{\partial x^2} < 0, \qquad \frac{\partial^2 P}{\partial x^2}\frac{\partial^2 P}{\partial y^2} - \frac{\partial^2 P}{\partial x\,\partial y}\frac{\partial^2 P}{\partial y\,\partial x} > 0$$

Endoodwracalny silnik Carnota

Rozwiązanie

$$\frac{x_0}{T_h} = \frac{1 - \sqrt{\frac{T_c}{T_h}}}{1 + \sqrt{\frac{\alpha_h}{\alpha_c}}}, \qquad \frac{y_0}{T_c} = \frac{\sqrt{\frac{T_h}{T_c}} - 1}{1 + \sqrt{\frac{\alpha_c}{\alpha_h}}}$$

Sprawność i moc w punkcie maksymalnej mocy

$$\eta = 1 - \frac{T_{cw}}{T_{hw}} = 1 - \sqrt{\frac{T_c}{T_h}},$$

$$P = \frac{\alpha_h \alpha_c}{\zeta} \left(\frac{\sqrt{T_h} - \sqrt{T_c}}{\sqrt{\alpha_h} + \sqrt{\alpha_c}} \right)^2$$

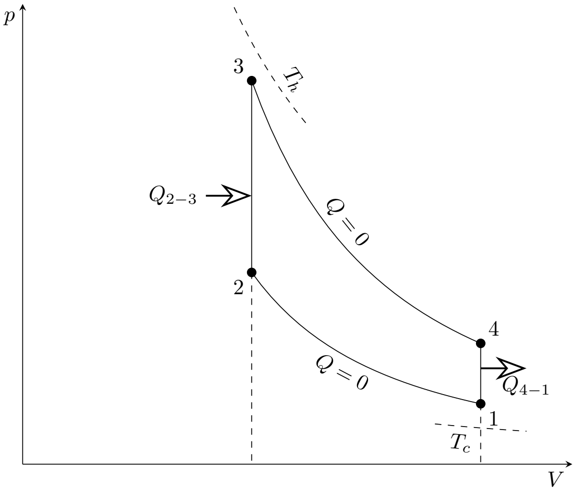
Endoodwracalny silnik Otto

Zmiana temperatury

$$\frac{\mathrm{d}T}{\mathrm{d}t} = -\alpha_h(T - T_h),$$
$$T(t=0) = T_2,$$
$$T(t=\tau_h) = T_3$$

$$\frac{\mathrm{d}T}{\mathrm{d}t} = -\alpha_c(T - T_c),$$
$$T(t=0) = T_4,$$
$$T(t=\tau_h) = T_1$$

$$\frac{T_1}{T_2} = \frac{T_4}{T_3} = \kappa$$



Schemat silnika Otto

Endoodwracalny silnik Otto

Temperatury

$$T_1 = \frac{T_c e^{\alpha_h \tau_h}(e^{\alpha_c \tau_c} - 1) + \kappa T_h (e^{\alpha_h \tau_h} - 1)}{e^{\alpha_h \tau_h + \alpha_c \tau_c} - 1},$$

$$T_2 = \frac{T_c e^{\alpha_h \tau_h}(e^{\alpha_c \tau_c} - 1) + \kappa T_h (e^{\alpha_h \tau_h} - 1)}{\kappa (e^{\alpha_h \tau_h + \alpha_c \tau_c} - 1)},$$

$$T_3 = \frac{T_c (e^{\alpha_c \tau_c} - 1) + \kappa T_h e^{\alpha_c \tau_c}(e^{\alpha_h \tau_h} - 1)}{\kappa (e^{\alpha_h \tau_h + \alpha_c \tau_c} - 1)},$$

$$T_4 = \frac{T_c (e^{\alpha_c \tau_c} - 1) + \kappa T_h e^{\alpha_c \tau_c}(e^{\alpha_h \tau_h} - 1)}{e^{\alpha_h \tau_h + \alpha_c \tau_c} - 1}$$

Endoodwracalny silnik Otto

Ciepła

$$Q_{2-3} = nc_V(T_3 - T_2) = nc_V \frac{(\kappa T_h - T_c)(e^{\alpha_h \tau_h} - 1)(e^{\alpha_c \tau_c} - 1)}{\kappa(e^{\alpha_h \tau_h - \alpha_c \tau_c} - 1)},$$

$$Q_{4-1} = nc_V(T_1 - T_4) = -nc_V \frac{(\kappa T_h - T_c)(e^{\alpha_h \tau_h} - 1)(e^{\alpha_c \tau_c} - 1)}{e^{\alpha_h \tau_h - \alpha_c \tau_c} - 1}$$

Moc

$$P = \frac{nc_V}{\zeta(\tau_h + \tau_c)} \frac{(1-\kappa)(\kappa T_h - T_c)}{\kappa} \sinh\left(\frac{\alpha_h \tau_h}{2}\right) \sinh\left(\frac{\alpha_c \tau_c}{2}\right) \cosh\left(\frac{\alpha_h \tau_h + \alpha_c \tau_c}{2}\right)$$

Sprawność

$$\eta = 1 - \kappa = 1 - \sqrt{\frac{T_c}{T_h}}$$

Endoodwracalny silnik Stirlinga

Ciepła

$$Q_{1-2} = nRT_{cw} \ln\left(\frac{V_2}{V_1}\right),$$

$$Q_{3-4} = nRT_{hw} \ln\left(\frac{V_1}{V_2}\right)$$

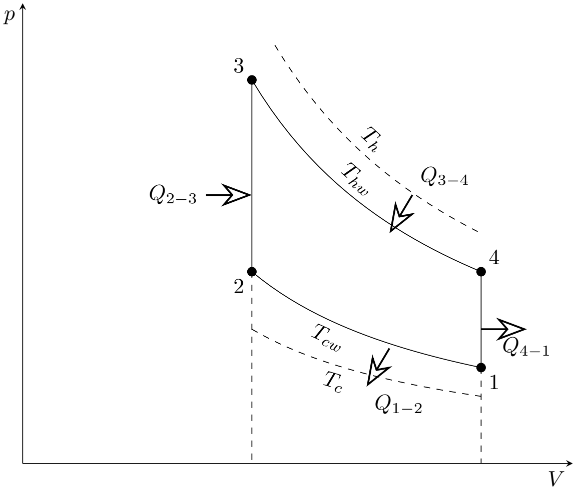
Entropia

$$\frac{Q_{1-2}}{T_{cw}} + \frac{Q_{3-4}}{T_{hw}} = 0,$$

$$\Delta S_{2-3} + \Delta S_{4-1} = 0$$

Sprawność

$$\eta = 1 - \frac{T_{cw}}{T_{hw}} = 1 - \sqrt{\frac{T_c}{T_h}}$$



Schemat silnika Stirlinga

Oscylator harmoniczny

Klasyczny hamiltonian

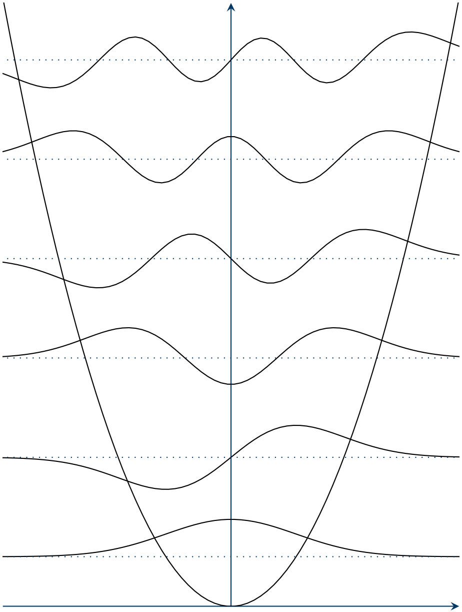
$$\mathcal{H} = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 q^2$$

Kwantowy hamiltonian

$$\hat{\mathcal{H}} = \frac{\hat{\boldsymbol{p}}^2}{2m} + \frac{m\omega^2}{2}\hat{\boldsymbol{q}}^2 = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial q^2} + \frac{m\omega^2}{2}q^2$$

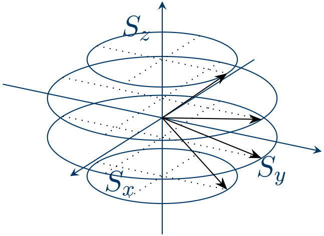
Energia

$$E_n = \hbar\omega\left(n + \frac{1}{2}\right)$$

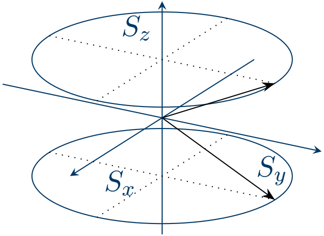


Układ o spinie $\frac{1}{2}$

Zagadnienie własne	Hamiltonian	Energia
$\hat{\boldsymbol{S}}^2 \lvert s m_s\rangle = \hbar^2 s(s+1)\lvert s m_s\rangle,$ $\hat{\boldsymbol{S}}_z \lvert s m_s\rangle = \hbar m_s \lvert s m_s\rangle$	$\hat{\mathcal{H}} = -\hat{\boldsymbol{\mu}} \cdot \boldsymbol{B} = -\gamma \hat{\boldsymbol{S}} \cdot \boldsymbol{B}$	$E_\uparrow = -\frac{\hbar\omega}{2}, \quad E_\downarrow = \frac{\hbar\omega}{2}$

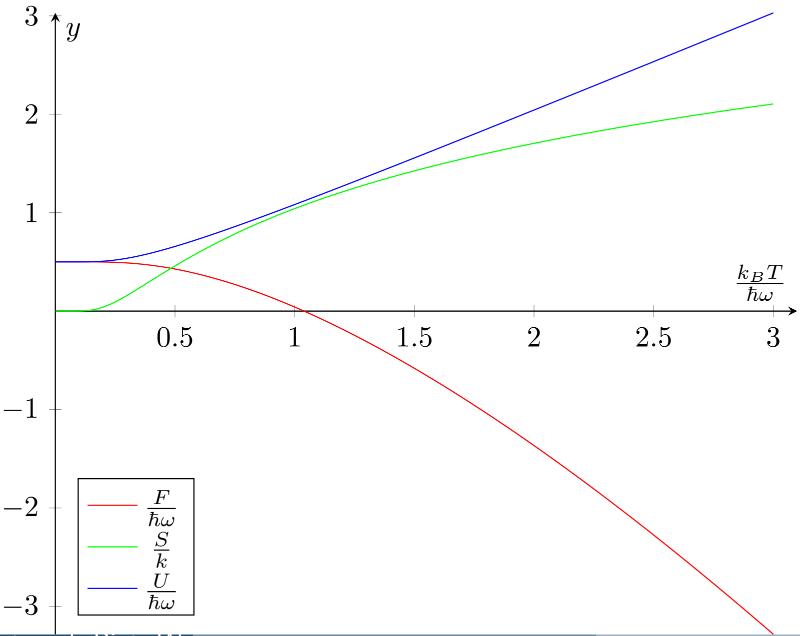


Stany dla $s = \frac{3}{2}$

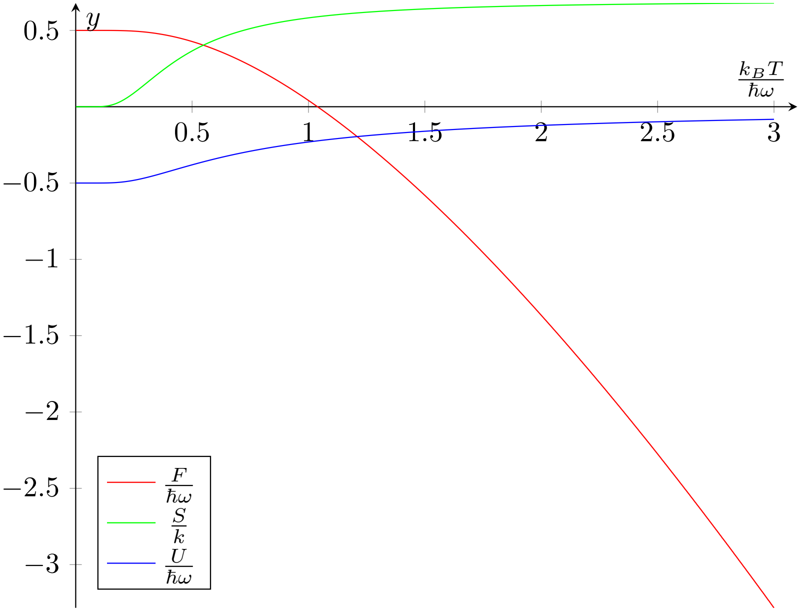


Stany dla $s = \frac{1}{2}$

Oscylator harmoniczny	Układ o spinie $\frac{1}{2}$
$p_n = \frac{1}{Z}\exp\left(-\frac{E_n}{k_B T}\right),$	$p_{m_s} = \frac{1}{Z}\exp\left(-\frac{E_{m_s}}{k_B T}\right),$
$Z = \frac{\exp\left(-\frac{\hbar\omega}{2k_B T}\right)}{1-\exp\left(-\frac{\hbar\omega}{2k_B T}\right)},$	$Z = 2\cosh\left(\frac{\hbar\omega}{2k_B T}\right),$
$U = \frac{\hbar\omega}{2}\,\mathrm{ctgh}\left(\frac{\hbar\omega}{2k_B T}\right),$	$U = -\frac{\hbar\omega}{2}\,\mathrm{tgh}\left(\frac{\hbar\omega}{2k_B T}\right),$
$S = \frac{\hbar\omega}{2T}\,\mathrm{ctgh}\left(\frac{\hbar\omega}{2k_B T}\right) - k_B \ln\left\{2\sinh\left(\frac{\hbar\omega}{2k_B T}\right)\right\}$	$S = -\frac{\hbar\omega}{2T}\,\mathrm{tgh}\left(\frac{\hbar\omega}{2k_B T}\right) + k_B \ln\left\{2\cosh\left(\frac{\hbar\omega}{2k_B T}\right)\right\}$


y
3
2
1
−1
−2
−3
0.5
1
1.5
2
2.5
3
$\frac{k_B T}{\hbar\omega}$
$\frac{F}{\hbar\omega}$
$\frac{S}{k}$
$\frac{U}{\hbar\omega}$

Układ o spinie $\frac{1}{2}$



Ciepła przemian

$$Q_{1-2} = 0, \qquad Q_{2-3} = \frac{\hbar\omega_2}{2}\left\{\mathrm{ctgh}\left(\frac{\hbar\omega_2}{2k_B T_3}\right) - \mathrm{ctgh}\left(\frac{\hbar\omega_2}{2k_B T_2}\right)\right\},$$

$$Q_{3-4} = 0, \qquad Q_{4-1} = \frac{\hbar\omega_1}{2}\left\{\mathrm{ctgh}\left(\frac{\hbar\omega_1}{2k_B T_1}\right) - \mathrm{ctgh}\left(\frac{\hbar\omega_1}{2k_B T_4}\right)\right\}.$$

Praca

$$-W = \frac{\hbar\omega_2}{2}(1-\kappa)\left\{\mathrm{ctgh}\left(\frac{\hbar\omega_2}{2k_B T_3}\right) - \mathrm{ctgh}\left(\frac{\hbar\omega_2}{2k_B T_2}\right)\right\}.$$

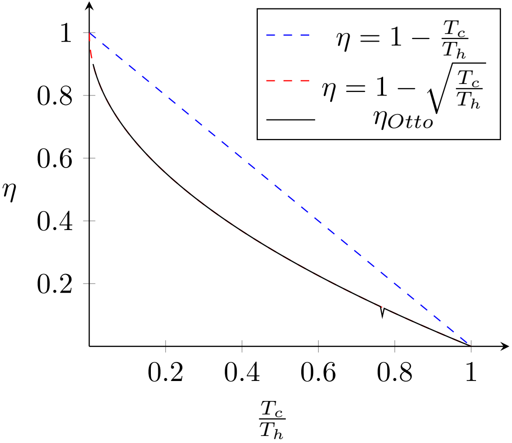
Silnik Otto z osc. harm.

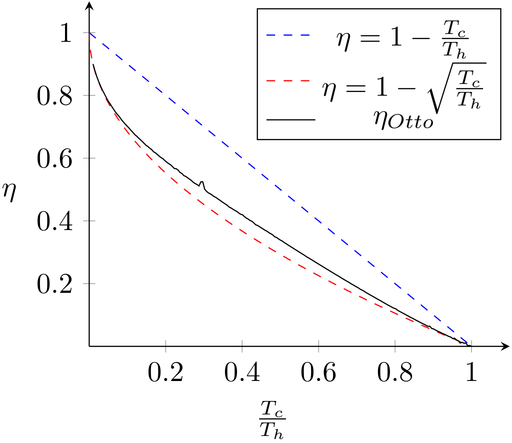
Sprawność

$$\eta = \frac{-W}{Q_{2-3}} = 1 - \kappa = 1 - \frac{\omega_1}{\omega_2}$$

Moc

$$P = \frac{\hbar\omega_2(1-\kappa)}{2\zeta(\tau_h+\tau_c)}\left[\mathrm{ctgh}\left(\frac{\hbar\omega_2}{2k_BT_3}\right) - \mathrm{ctgh}\left(\frac{\hbar\omega_2}{2k_BT_2}\right)\right]$$





Ciepła przemian

$$Q_{1-2} = 0, \qquad Q_{2-3} = \frac{\hbar\omega_2}{2}\left\{\mathrm{tgh}\left(\frac{\hbar\omega_2}{2k_BT_2}\right) - \mathrm{tgh}\left(\frac{\hbar\omega_2}{2k_BT_3}\right)\right\},$$

$$Q_{3-4} = 0, \qquad Q_{4-1} = \frac{\hbar\omega_1}{2}\left\{\mathrm{tgh}\left(\frac{\hbar\omega_1}{2k_BT_4}\right) - \mathrm{tgh}\left(\frac{\hbar\omega_1}{2k_BT_1}\right)\right\}.$$

Praca

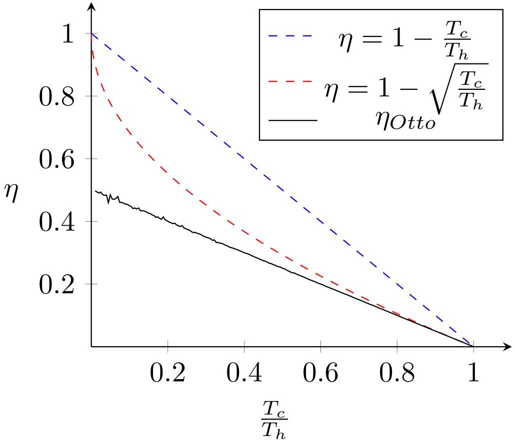
$$-W = \frac{\hbar\omega_2}{2}(1-\kappa)\left\{\mathrm{tgh}\left(\frac{\hbar\omega_2}{2k_BT_2}\right) - \mathrm{tgh}\left(\frac{\hbar\omega_2}{2k_BT_3}\right)\right\}.$$

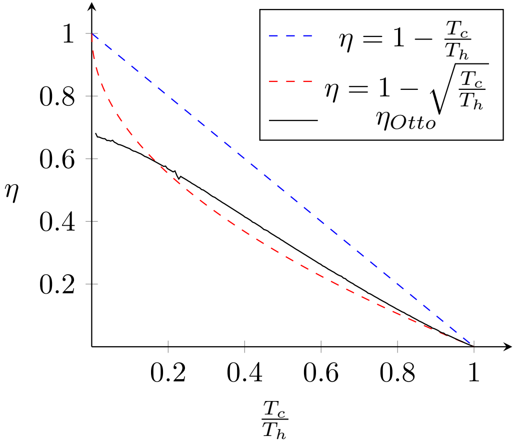
Sprawność

$$\eta = \frac{-W}{Q_{2-3}} = 1 - \kappa = 1 - \frac{\omega_1}{\omega_2}$$

Moc

$$P = \frac{\hbar\omega_2(1-\kappa)}{2\zeta(\tau_h+\tau_c)}\left[\mathrm{tgh}\left(\frac{\hbar\omega_2}{2k_BT_2}\right) - \mathrm{tgh}\left(\frac{\hbar\omega_2}{2k_BT_3}\right)\right]$$





Ciepła przemian

$$Q_{1-2} = k_B T_c \ln\left[2\sinh\left(\frac{\hbar\omega_1}{2k_B T_c}\right)\right] - k_B T_c \ln\left[2\sinh\left(\frac{\hbar\omega_2}{2k_B T_c}\right)\right] + \\ + \frac{\hbar\omega_2}{2}\operatorname{ctgh}\left(\frac{\hbar\omega_2}{2k_B T_c}\right) - \frac{\hbar\omega_1}{2}\operatorname{ctgh}\left(\frac{\hbar\omega_1}{2k_B T_c}\right),$$

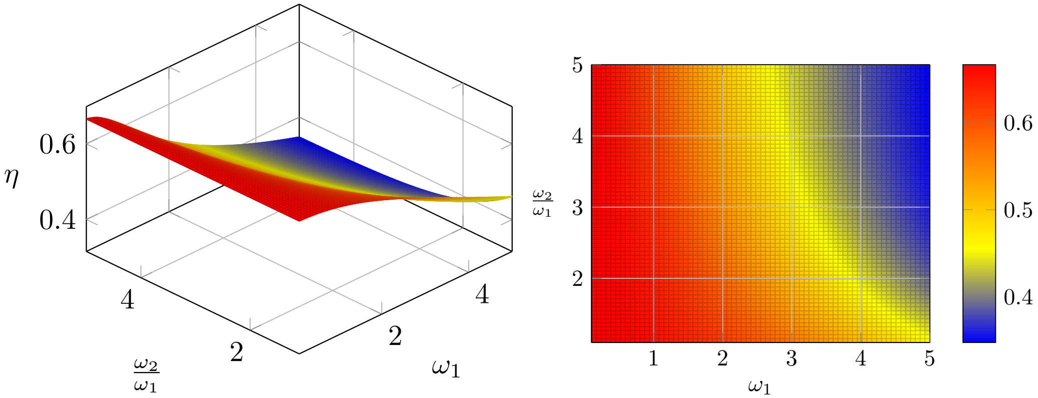
$$Q_{2-3} = \frac{\hbar\omega_2}{2}\operatorname{ctgh}\left(\frac{\hbar\omega_2}{2k_B T_h}\right) - \frac{\hbar\omega_2}{2}\operatorname{ctgh}\left(\frac{\hbar\omega_2}{2k_B T_c}\right),$$

$$Q_{3-4} = k_B T_h \ln\left[2\sinh\left(\frac{\hbar\omega_2}{2k_B T_h}\right)\right] - k_B T_h \ln\left[2\sinh\left(\frac{\hbar\omega_1}{2k_B T_h}\right)\right] + \\ + \frac{\hbar\omega_1}{2}\operatorname{ctgh}\left(\frac{\hbar\omega_1}{2k_B T_h}\right) - \frac{\hbar\omega_2}{2}\operatorname{ctgh}\left(\frac{\hbar\omega_2}{2k_B T_h}\right),$$

$$Q_{4-1} = \frac{\hbar\omega_1}{2}\operatorname{ctgh}\left(\frac{\hbar\omega_1}{2k_B T_c}\right) - \frac{\hbar\omega_1}{2}\operatorname{ctgh}\left(\frac{\hbar\omega_1}{2k_B T_h}\right)$$

Praca

$$-W = k_B T_h \ln\left[\frac{\sinh\left(\frac{\hbar\omega_2}{2k_B T_h}\right)}{\sinh\left(\frac{\hbar\omega_1}{2k_B T_h}\right)}\right] + k_B T_c \ln\left[\frac{\sinh\left(\frac{\hbar\omega_1}{2k_B T_c}\right)}{\sinh\left(\frac{\hbar\omega_2}{2k_B T_c}\right)}\right]$$

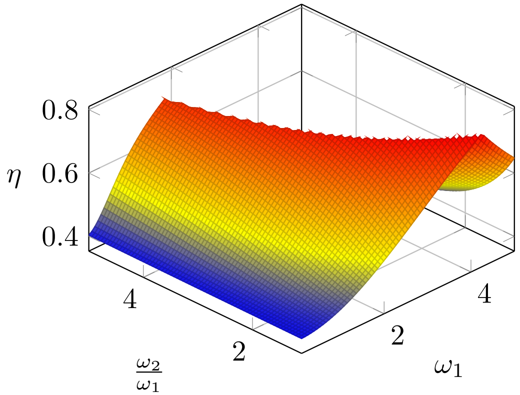


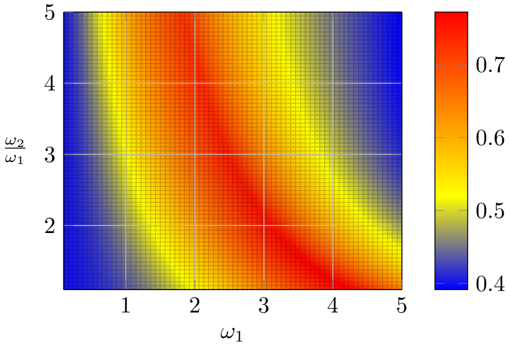
Ciepła przemian

$$Q_{1-2} = k_B T_c \ln\left[2\cosh\left(\frac{\hbar\omega_2}{2k_B T_c}\right)\right] - k_B T_c \ln\left[2\cosh\left(\frac{\hbar\omega_1}{2k_B T_c}\right)\right] + $$
$$+ \frac{\hbar\omega_1}{2}\,\mathrm{tgh}\left(\frac{\hbar\omega_1}{2k_B T_c}\right) - \frac{\hbar\omega_2}{2}\,\mathrm{tgh}\left(\frac{\hbar\omega_2}{2k_B T_c}\right),$$
$$Q_{2-3} = \frac{\hbar\omega_2}{2}\,\mathrm{tgh}\left(\frac{\hbar\omega_2}{2k_B T_c}\right) - \frac{\hbar\omega_2}{2}\,\mathrm{tgh}\left(\frac{\hbar\omega_2}{2k_B T_h}\right),$$
$$Q_{3-4} = k_B T_h \ln\left[2\cosh\left(\frac{\hbar\omega_1}{2k_B T_h}\right)\right] - k_B T_h \ln\left[2\cosh\left(\frac{\hbar\omega_2}{2k_B T_h}\right)\right] + $$
$$+ \frac{\hbar\omega_2}{2}\,\mathrm{tgh}\left(\frac{\hbar\omega_2}{2k_B T_h}\right) - \frac{\hbar\omega_1}{2}\,\mathrm{tgh}\left(\frac{\hbar\omega_1}{2k_B T_h}\right),$$
$$Q_{4-1} = \frac{\hbar\omega_1}{2}\,\mathrm{tgh}\left(\frac{\hbar\omega_1}{2k_B T_h}\right) - \frac{\hbar\omega_1}{2}\,\mathrm{tgh}\left(\frac{\hbar\omega_1}{2k_B T_c}\right)$$

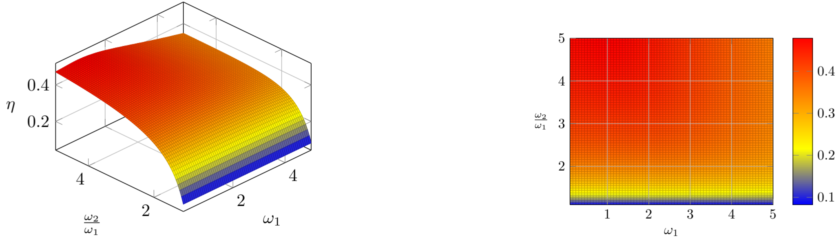
Praca

$$-W = k_B T_h \ln\left[\frac{\cosh\left(\frac{\hbar\omega_1}{2k_B T_h}\right)}{\cosh\left(\frac{\hbar\omega_2}{2k_B T_h}\right)}\right] + k_B T_c \ln\left[\frac{\cosh\left(\frac{\hbar\omega_2}{2k_B T_c}\right)}{\cosh\left(\frac{\hbar\omega_1}{2k_B T_c}\right)}\right]$$

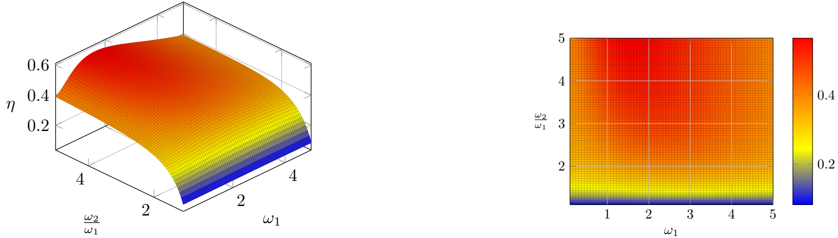




Silnik Stirlinga bez regeneratora



Rysunek: Silnik Stirlinga z oscylatorem harmonicznym



Rysunek: Silnik Stirlinga ze spinem $\frac{1}{2}$

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